

Predictive analytics for loan default in banking sector using machine learning techniques

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Abstract— The early prediction of loan default in banks is a critical area nowadays. Machine learning techniques are being used to predict and classify the loan applicant as defaulter or not before granting him the loan. Several types of statistical and artificial intelligent classification techniques can be used to predict the loan default of bank's customers. However, ensemble machine learning techniques are introduced to perform better than traditional statistical and AI techniques as it combines several base models to produce one optimal predictive model with better performance and accuracy. A comparative study is applied on different machine learning techniques on customer's loans dataset obtained from a public bank's database that contains customer's loans and personal data. The data is processed and analyzed using Apache Spark, a machine learning tool for big data processing. The results of the proposed system is evaluated by seven performance metrics to compare the performance of each classifier and find out the best performing one among them. It is found out that the ensemble machine learning techniques has better performance than single base classifiers in predicting the loan default.

Keywords—loan default, machine learning, ensemble learning, predictive model, classification

I. INTRODUCTION

One of the most risky processes that the banks face nowadays is loan evaluation problem because of the severe impact of loan default. This process requires collecting large amount of credit data to be analyzed and classified to be able to assess the credit situation of the applicant and take the right decision of whether granting or refusing the loan application.

Predictive analytics is one of the fields of data mining that extract and analyze hidden information from raw data to predict customer's behaviors. For example, identifying fraudulent behavior of credit cards. Predictive analytics apply several techniques to be able to learn from historical data and predict possible future outcomes.[1]

Credit scoring models objective is to assess the credit or loan applicant as good or bad to determine his repayment behavior whether he is likely to be defaulter or not. Thus, credit scoring is generally exposed to classification problems. [2]

Classifying customer's data became a big problem, as the information in the databases became larger. Traditional statistical methods such as Discriminant Analysis and Logistic Models has been evolved to non-parametric techniques as Neural Networks and Support Vector Machines. However recently, machine learning became a vital alternative to solve binary classification problems [3].

Machine learning is an artificial intelligence technology used to learn from given data and apply the learnt rules on new data without any intervention from human or lots of programming. ML helps to make decisions or predictions by analyzing the given data and learning from it then works on improving the knowledge by continuous learning from the out coming results. Machine learning methods are classified into three types, supervised, semi-supervised and unsupervised learning [4].

There is many complicated techniques to be used nowadays for enhancing the credit scoring prediction. One of them is combining more than one classifier which is known as ensemble machine learning that produce one optimal model with enhanced performance [5]. In this paper, we used various classification techniques and compared their performance with ensemble learning techniques to assess the customers as loan defaulters or not and find out the best performing classifier.

The rest of the paper is organized as follows: Sections 2 presents the proposed system. Section 3 describes the customer's loans dataset. Section 4 describes the experimental results and discussing them. Finally, Section 5 describes the conclusion and future work.

II. BACKGROUND

A. Loan default in banking

Predictive analytics is aiming to predict if the customer will be creditworthy or not and what are the promotions that he may respond to, using statistical and machine learning techniques that are all part of data mining. Predictive analytical applications use different techniques including classification, clustering and association algorithms. [6]

It is stated by Hand and Jacka (1998) that assessing the creditworthiness of customer is called to be credit scoring. Credit scoring is the decision modelling and using data mining techniques that aid banks in assessing the decision of granting loans to the customers. [7]

B. Techniques used for loan default classification

Evolution of the techniques used for credit scoring occurred, ranging from expert scoring models to statistical models as LDA, Logistic regression and Bayesian model. Because of the development of the machine learning, artificial intelligence models appeared as they have higher accuracy than statistical models such as artificial neural networks (ANN) and support vector machines (SVM). Hybrid & ensemble machine learning techniques are the best performing nowadays. Hybrid learning uses one classifier for training sample and uses others in feature selection and classifying

stages. While ensemble learning methods uses different classifiers with different parameters and train them on different samples more than once. [8]

1. Statistical techniques

- **K-nearest neighbor:** K-NN is a classification technique which assign the new observation's class according to distance function between it and the nearest neighbors by majority voting where the new observation is assigned to the most common class among its k neighbors. [9] The distance between two samples is defined in terms of Euclidean distance: [8]

$$d(X,Y) = \sqrt{\sum_{i=1}^n (X_i - Y_i)^2} \quad (1)$$

- **Logistic regression:** it is a classification function that put boundaries between classes and determine the class probability according to the distance from this boundary where this probability may be classified into 0 or 1 to make classification easier and more detailed. [10]

The Regression equation is presented as follows [11]: $\ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \varepsilon_i$ (2)

Where p_i represents the probability of being "good" for a particular customer i which is also a function of the predictive variables X_i , the variables of applicant's characteristics as age, loan amount, occupancy,...

β_0 is the intercept, $\beta_j = (1, \dots, 4)$ represent the coefficients associated with the corresponding predictor X_i ($i = 1, \dots, 4$); $\ln\left(\frac{p_i}{1-p_i}\right)$ represents the default event (Y_i), and ε_i is the errors' term.

- **Decision Trees:** Decision Trees (DT) classify instances represented as a tree where the feature of an instance is represented as a node and the branch of the node represents the feature value. Features are classified according to the feature values. [10]

2. Artificial intelligence methods

- **Neural Networks (NN):** this technique is driven by mental learning and neuro biological systems. It consists of interconnected neurons that works independent in parallel. These neurons learn from stored information and the messages are carried by the interconnections between the neurons. The strength of the signals differs according to the weight of the connections between neuron [12]. NN have higher prediction accuracy than statistical methods (Odom & Sharda, 1990; Abdou et al., 2008). The number of neurons and layers depends on the complexity of the system being studied. The neurons in the input layer receive

and transfer the data to neurons in the hidden layers through the weighted links for processing, and the result is transferred to the neurons in the output layer for analysis. [3][11]

The hidden layer is presented in the following equations:

$$u_k = \sum w_{ki} x_i - \theta_k \quad (3)$$

$$H_k = f(u_k) \quad (4)$$

And the output layer is presented in the following equations

$$u_j = \sum w_{jk} H_k - \theta_j \quad (5)$$

$$Y_j = f(u_j) \quad (6)$$

Where , u_k is the input for hidden unit k , H_k is the output of the hidden unit k , x_i is the value of input variable i , w_{ki} is the weight giving the intensity of the signal given by input variable i to the hidden neuron k , θ_k is the threshold value, $f(u_k)$ is the activation function for the hidden unit k , u_j is the input for output unit j , w_{jk} is the link from hidden neuron k to output layer j , θ_j is the threshold value, Y_j is the output for the final layer and $f(u_j)$ is the activation function for the output unit j . [3]

3. Ensemble methods

- **Random Forest:** it is ensemble of random decision trees, that combines the predictions of random selected individual trees. [13] The class of the instance is usually predicted either by averaging the predictions of all trees in case of regression or as the class with the majority vote for classification. [14]
- **Gradient Boosted Decision Trees (GBDT):** it is ensemble-learning technique of decision trees (Friedman, 2001). It works on enhancing the classification performance of the decision trees by reducing the training error through moving against the gradient's direction so that it can minimize the loss function, converting the weak learners into strong ones. [15]

III. PROPOSED SYSTEM

The proposed system is for early prediction of bank's customers loan default based on machine learning algorithms using statistical, AI and ensemble learning techniques. The model is built using the variables from the data set provided by Egyptian public sector bank. This data is used to develop the best performing prediction model for future customer evaluation.

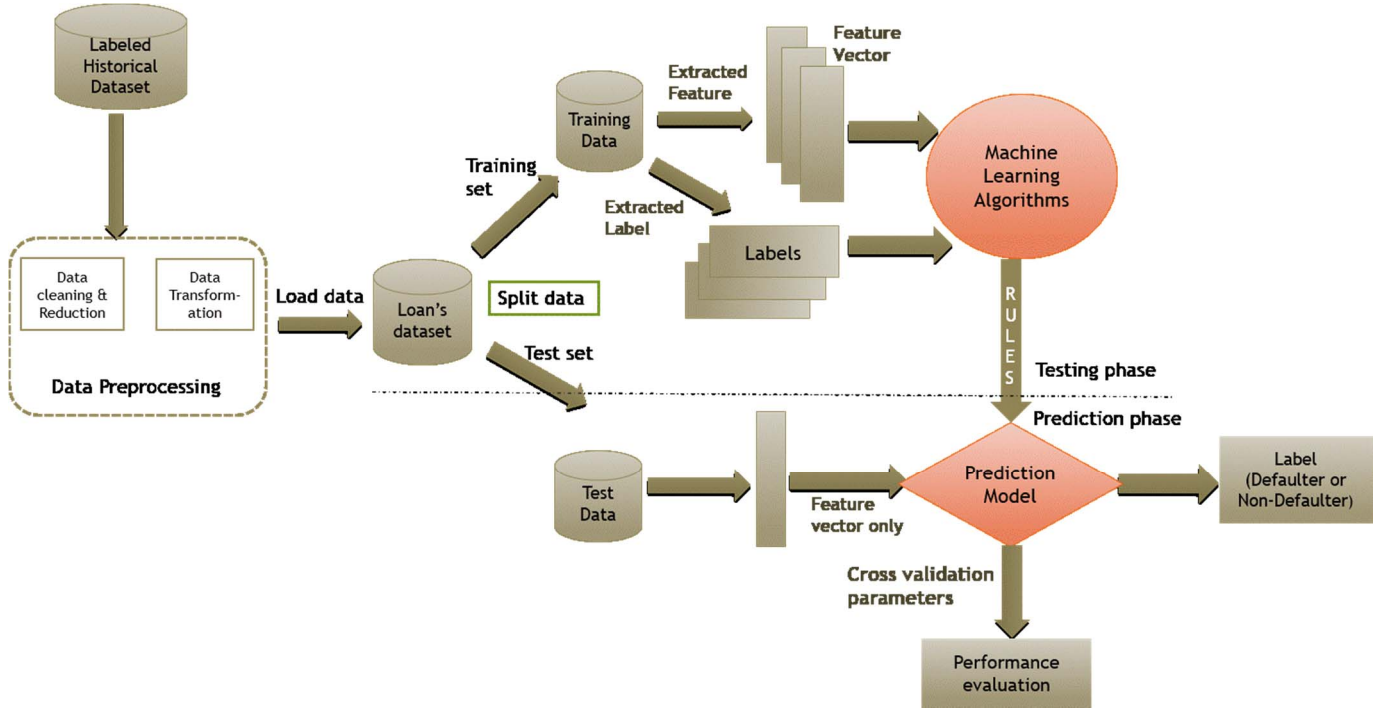


Fig.1. The proposed model

The model is composed of the following stages: (1) Data understanding (2) Data preparation (3) Data loading (4) Modelling (5) Evaluation as shown in “Fig. 1”

A. Data Understanding

The data set is consisting of 2,954,168 observations representing the loans borrowed by the bank’s customers over the previous 13 years since May 2005. The model is based on supervised learning where the customer’s data are labeled into defaulter and non-defaulter.

B. Data Preparation

The dataset undergoes some preprocessing steps in order to remove the redundant attributes, rescale the observations and handle the extreme values of the variables to be transformed into a single form understandable by the machine learning tool used. The steps includes:

- Data cleansing & reduction
- Feature standardization
- Feature normalization
- Feature engineering

C. Data Loading

The third step includes concatenating the data in one sheet and loading the transformed data into a single database all with the same format (CSV format) so that this data can be used to apply machine learning techniques on.

D. Classification based on machine learning algorithms

After preparing the data, it has been uploaded to the machine learning tool to apply the ML algorithms. The machine learning tool used is Apache Spark.

The first step includes employing the machine learning algorithms on this model. From the statistical algorithms, we selected decision tree and logistic regression, from artificial intelligent algorithms selected the artificial neural networks.

The feed forward artificial neural network applied consists of number of layers of nodes. Each layer is interconnected with the neighbor layer. The input layer represents the input data. Inputs are mapped into outputs through a linear combination between input nodes with weight w and bias b which is presented by this equation:

$$y(x) = f_k(\dots f_2(w_2^T f_1(w_1^T x_1 + b_1) + b_2) \dots + b_k) \quad (7)$$

Nodes in intermediate layers use sigmoid (logistic) function:

$$f(z_i) = \frac{1}{1+e^{-z_i}} \quad (8)$$

Nodes in the output layer use softmax function:

$$f(z_i) = \frac{e^{z_i}}{\sum_{k=1}^N e^{z_k}} \quad (9)$$

The number of classes is represented by the nodes N in the output layer

However, from the ensemble learning techniques, we selected random forests and gradient boosted trees.

The second step includes computing the classification results of each algorithm and creating a comparative study between the results obtained using seven performance metrics to get the highest performing algorithm among them.

The third part includes computing the best classifier for the data set obtained according to the comparative study performed to improve the model’s prediction accuracy and precision.

1. Modelling

For the modelling phase, the dataset is split into training, validation and test sets. Cross validation begins by splitting the dataset into a set of folds which are used as separate training and test datasets. In our case the models were trained using $k=10$ folds, hence cross validation will generate 10 (training, test) dataset pairs, each of which uses 9/10 of the data for training and 1/10 for testing.

Training data set is used to build and train the predictive model. As we are working on supervised learning, we selected the mostly common used classification learning techniques based on the literature review done such as KNN, Decision Trees, Logistic Regression, Neural Networks and Ensembles e.g. Random Forests and Gradient Boosted Trees. These techniques are implemented to develop a model which predicts whether a loan will be paid back from a bank's customer or not.

We validated the six trained predictive models using validation data set. The validation data set was also used for model fine-tuning and to check the over-fitting of models. The confusion matrix is provided for each model respectively to evaluate the performance of each model on validation data set.

2. Metrics of Evaluation

The performance metrics are used to evaluate the classifiers used. We used the following performance metrics: [4]

- Confusion Matrix: It contains information about predicted classified instances (positive and negative) against actual classifications (positive and negative).
- Accuracy: the proportion of correctly classified cases to all cases in the set. The larger the predictive accuracy the better the situation.
Accuracy = $TP+TN/TP+TN+FP+FN$ (10)
- Precision: It is a measure which estimates the probability that a positive predicted value is correct. Precision is calculated by using the formula: Precision = $TP/TP+FP$ (11)
- Sensitivity: it is the true positive rate, it measures the probability of test outcome to be positive when it is actually positive.
Sensitivity = $TPR = TP/P = TP/(TP+FN)$ (12)
- Specificity: it is the true negative rate, measures the probability of test outcome to be negative when it is actually negative.
= $TNR = TN/N = TN/(TN+FP) = 1-FPR$ (13)
- Fall out: it is the false positive rate which is the proportion of those predicted to be positive but they are actually negative.
= $FP/N = FP/FP+TN = 1-TNR$ (14)
- Miss rate: it is the false negative rate which is the proportion of those predicted to be negative but they are actually positive.
= $FN/N = FN/FN+TP$ (15)

IV. DATASET DESCRIPTION

Dataset is collection of loan's data of bank customers. The dataset used for this experiment consists of 2,954,168 observations from a public Egyptian bank's database. These instances contain data about the customer's personal, demographic and acquired loan's data. The data collection period was from May 2005 till December 2017. This big data set is collected from the bank's database anonymously taking into consideration the customer's data privacy. The file format of the dataset used is Comma Separated Value CSV to be directly uploaded to the machine learning tool. The ML tool used is Apache Spark applied on cloud server with 32 GB RAM, 8 cores and storage 28 GB. The description of the attributes is shown in Table I.

TABLE I. DATA DESCRIPTION

Attribute	Description
ID	Identification number of the customer
Age	Age
Gender	Male/Female
Marital status	Married/Single
Address	State
Loan type	Personal loan, car loan, etc..
Amount disbursement	Amount of loan taken
Date Disbursement	Data of loan acquisition
Data maturity	Data of installments payment
Balance Book	Total amount of unpaid payments
Installment amount	Amount of each installment to be paid
Total amount arrears	Amount of late payments
Number of installment arrear	Number of late monthly installments
label	Defaulter or non-defaulter

V. EXPERIMENTAL RESULTS

The following results present the experiment applied on customer's loans dataset within the Apache Spark platform to evaluate the performance of the six selected classification algorithms according to the seven performance metrics. Then the dataset is divided into training and testing for modeling with ratio 70:30. Comparison is applied on the selected classifiers: K-nearest neighbor, decision trees, logistic regression, neural networks, random forest and gradient boosted trees. The confusion matrix of each classifier is shown in Table II, Table III, Table IV, Table V, Table VI and Table VII respectively.

TABLE II. CONFUSION MATRIX RESULTS OF K-NEAREST NEIGHBOR CLASSIFIER

Confusion Matrix		Predicted	
		0	1
Actual	0	True Positive	False Negative
		6341	7913
	1	False Positive	True Negative
		69464	803045

TABLE III. CONFUSION MATRIX RESULTS OF DECISION TREES CLASSIFIER

Confusion Matrix		Predicted	
		0	1
Actual	0	True Positive	False Negative
		9653	19677
	1	False Positive	True Negative
		66152	791281

TABLE IV. CONFUSION MATRIX RESULTS OF LOGISTIC REGRESSION CLASSIFIER

Confusion Matrix		Predicted	
		0	1
Actual	0	True Positive	False Negative
		480	402
	1	False Positive	True Negative
		75325	810556

TABLE V. CONFUSION MATRIX RESULTS OF NEURAL NETWORK CLASSIFIER

Confusion Matrix		Predicted	
		0	1
Actual	0	True Positive	False Negative
		0	0
	1	False Positive	True Negative
		75805	810958

TABLE VI. CONFUSION MATRIX RESULTS OF GRADIENT BOOSTED TREES CLASSIFIER

Confusion Matrix		Predicted	
		0	1
Actual	0	True Positive	False Negative
		8140	3991
	1	False Positive	True Negative
		67665	806967

TABLE VII. CONFUSION MATRIX RESULTS OF RANDOM FOREST CLASSIFIER

Confusion Matrix		Predicted	
		0	1
Actual	0	True Positive	False Negative
		2837	2
	1	False Positive	True Negative
		72968	810956

TABLE VIII. COMPARATIVE SUMMARY OF THE CLASSIFIERS PERFORMANCE METRICS

Classifier name	Accuracy	Precision	Sensitivity (TP rate)	Specificity (TN rate)	Fall out (FP rate)	Miss rate (FN rate)
Decision tree	90.3%	62.59 %	32.91 %	92.28%	7.7%	67.1%
K-nearest neighbor	91.2%	68.26 %	44.48 %	92.03%	7.9%	55.5%
Logistic Regression	91.4%	72.95 %	54.42 %	91.49%	8.5%	45.5%
Neural network	91.4%	45.72 %	0 %	91.45%	8.5%	0%
Random forests	91.7%	95.83%	99.9%	91.74%	8.3%	0%
Gradient Boosted Trees	91.9%	79.68%	67.10%	92.26%	7.7%	32.9%

Table VIII shows a comparative study between the performance metrics results of the six used classifiers. From the above table we see that the ensemble machine learning techniques- Random Forest & Gradient Boosted Trees- got better results rather than the rest of the models. RF achieved accuracy 91.77% and precision 95.83% however GBT achieved accuracy 91.91% and precision 79.68%. This indicates that ensemble learning techniques performs better than statistical and artificial intelligent learning techniques as they achieved higher accuracy and precision rates.

Performance measures of ML classifiers

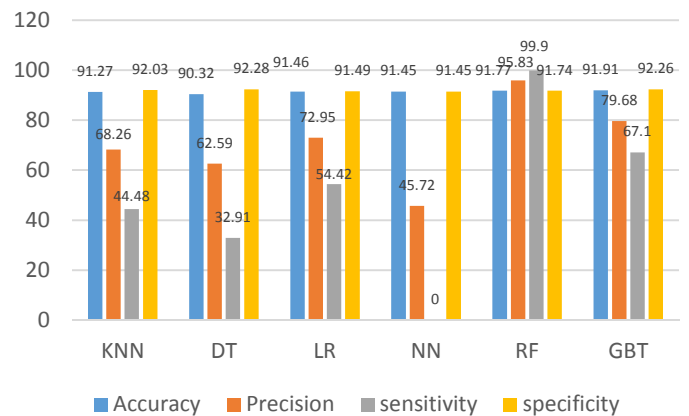


Fig.2. comparative study between classifiers as bar chart

In Fig “2” the comparative study performed is presented as bar chart where the X-axis represents the classifiers used and the Y-axis represents the performance metrics results ranging from 0 to 100%. The blue bar presents the accuracy, sensitivity is presented by grey bar, precision is presented by orange bar and the yellow bar presents the specificity. According to this comparison the best classifiers based on the performance metrics are Random Forest and Gradient Boosted Trees as they have the highest rates.

The results of the proposed model may vary according to the used dataset. Prediction is based on the used customer’s credit cards dataset which may vary from one bank to another according to the customer’s profiles, the bank’s policy in granting loans and the customer’s spending behavior. Thus, a detailed analysis should be applied on the customer’s dataset before applying the model.

As a contribution to the results obtained in this paper, the verified customer’s data set can further be used in customers marketing prediction model, to customize the bank’s products to the good customers classified to be “non-defaulter” as they will be profitable for the bank and widen the market range.

VI. CONCLUSION

In this paper, machine learning techniques have been applied on customer’s loans data from a public bank sector. Six different classification algorithms were applied on the dataset including statistical, artificial intelligence and ensemble learning methods. Seven performance metrics were used to evaluate the performance, comparing their results. It is

found that the highest performing models with the highest accuracy and precision rates for predicting the defaulter customer are Random Forest as it got 91.7% accuracy and 95.83% precision and Gradient Boosted Trees as it got 91.9% accuracy and 79.68% precision compared to K-NN, DT, LR, GBT and NN. However, among ensemble techniques, Random Forest got higher precision, sensitivity and specificity rates compared to Gradient Boosted Trees. These results show that ensemble learning techniques has better performance than single base classifiers.

In the future work, we will be using stream data instead of batch data like social media and customer calls data to enhance the learning process. This will need using deep learning techniques as they outperforms other techniques in voice, image and sentiment classification and analysis.

REFERENCES

- [1] Harsh Jain, Amrit Pal, Manish Kumar, “ Predictive Analytics in Operations Management”, Decision Management: Concepts, Methodologies, Tools, and Applications p.66, January 2017
- [2] Wenbing XIAO, Qian ZHAO, Qi FEI,” A Comparative Study of Data Mining Methods in Consumer Loans Credit Scoring Management”, Systems Engineering Society of China & Springer-Verlag 2006
- [3] Ion Smeureanu, Gheorghe Ruxanda, Laura Maria Badea,”CUSTOMER SEGMENTATION IN PRIVATE BANKING SECTOR USING MACHINE LEARNING TECHNIQUES”, Journal of Business Economics and Management Sept 2017
- [4] J. Sujatha, Dr. S.P. Rajagopalan,” Performance Evaluation of Machine Learning Algorithms in the Classification of Parkinson Disease Using Voice Attributes”, International Journal of Applied Engineering Research ISSN 0973-4562 Volume 12, Number 21 (2017) pp. 10669-10675
- [5] G. Arutjothi, Dr. C. Senthamarai,” A Prediction Model for Credit Defaulters using Ensemble Learning Classifier”, International Journal of Innovations & Advancement in Computer Science, October 2017
- [6] Ramesh Sharda, Dursun Delen, Efraim Turban, “Business Intelligence a managerial perspective on analytics third edition, P. 8”
- [7] HUSSEIN A. ABDOUa * AND JOHN POINTON, “CREDIT SCORING, STATISTICAL TECHNIQUES AND EVALUATION CRITERIA: A REVIEW OF THE LITERATURE”, Wiley Online Library (wileyonlinelibrary.com) DOI: 10.1002/isaf.325, Intell. Sys. Acc. Fin. Mgmt. 18, 59–88 (2011)
- [8] Xiao-Lin Li, Yu Zhong, “An Overview of Personal Credit Scoring: Techniques and Future Work”, International Journal of Intelligence Science, 2012, 2, 181-189
- [9] Mutua Jennifer Ndanu1, Gichuhi Anthony Waititu2, Wanjoya Anthony Kibera2, Muia Patricia Nthoki3,” Cluster Analysis, K-Nearest Neighbour and Artificial Neural Network Applied to Credit Data to Classify Credit Applicants”, American Journal of Theoretical and Applied Statistics 2016; 5(4): 186-191
- [10] Osisanwo F.Y., Akinsola J.E.T., Awodele O., Hinmikaiye J. O., Olakanmi O. ,Akinjobi J., Supervised Machine Learning Algorithms: Classification and Comparison, International Journal of Computer Trends and Technology (IJCTT) – Volume 48 Number 3 June 2017
- [11] Lobna Abid, Afif Masmoudi, Sonia Zouari-Ghorbel3,” THE CONSUMER LOAN’S PAYMENT DEFAULT PREDICTIVE MODEL: AN APPLICATION IN A TUNISIAN COMMERCIAL BANK”, Asian Economic and Financial Review, 2016, 6(1): 27-42
- [12] Dorota Witkowska, Wladyslaw Kaminski & Iwona Staniec,” THE LOAN GRANTING PROCEDURE: ARTIFICIAL NEURAL NETWORKS, DISCRIMINANT ANALYSIS, K-MEANS METHOD”, IFAC Modeling and Control of Economic Systems. Klagenfurt. Austria. 2001
- [13] Celine Vens, Fabrizio Costa,” Random Forest Based Feature Induction”, IEEE 2011
- [14] Jin Li*, Maggie Tran, Justy Siwabessy, “Selecting Optimal Random Forest Predictive Models: A Case Study on Predicting the Spatial Distribution of Seabed Hardness”, DOI:10.1371/journal.pone.0149089 February 18, 2016
- [15] Resul Tugay, Sule Gunduz Oguducu,” Demand Prediction Using Machine Learning Methods and Stacked Generalization”, july 2017R. Nicole, “Title of paper with only first word capitalized,” J. Name Stand. Abbrev., in press.